

What is Aletheon, in one sentence?

An AI-powered drug intelligence platform that aggregates evidence from FDA, clinical trials, PubMed, EMA, and real-world adverse event data, and produces a cited, tier-ranked research report in minutes — work that currently takes a pharmacist or researcher hours.

What problem are you solving, and who has this problem?

Pharma researchers, clinical pharmacists, medical writers, and drug safety teams spend hours manually cross-checking scattered sources. The cost of missed evidence can be patient safety incidents or regulatory non-compliance. Aletheon collapses this to minutes with source attribution.

What does "evidence tier" mean, and why does it matter?

Not all sources are equal. Regulatory labels (FDA/EMA) are ground truth. Peer-reviewed trials are authoritative. Real-world reports are signal but noisy. Preprints are early. Aletheon ranks every piece of evidence by this trust tier so the reader knows how much weight to give each finding — unlike a Google search that treats all sources identically.

Why now? Has this not been done before?

LLMs have only become capable of reliable evidence synthesis and citation in the past 2 years. Previous tools either gave raw search results (no synthesis) or gave synthesis without sources (hallucination risk). Aletheon combines both: structured multi-source fetch + LLM synthesis + source traceability.

1. Walk us through the pipeline end to end.

User inputs a drug name → parallel fetch from FDA labels, ClinicalTrials.gov, Europe PMC, FAERS → evidence normalized into a typed Evidence object with tier + source → Qdrant vector store for semantic retrieval → LangGraph orchestration (fetch → combine → index → retrieve → report → evaluate → correct → critic) → PDF report with tier-color-coded citations.

Hard

2.

How do you prevent hallucinations?

Three layers: (1) every claim in the report is grounded to a fetched Evidence object with a real source URL, (2) a self-evaluation loop checks the report against retrieved evidence and flags unsupported claims, (3) a critic node does a final pass. The system cannot cite a source it didn't actually retrieve.

Hard

3.

What if two sources contradict each other?

Higher-tier evidence takes precedence — regulatory label overrides a preprint. Contradictions within the same tier are surfaced explicitly in the report as

"conflicting findings," with both sources cited, so the reader can judge. The system doesn't silently resolve conflicts.

Probe

4.

How do you handle a drug with very little published data?

Each source fetch degrades gracefully — if a source returns nothing, it's logged as empty rather than crashing the pipeline. The report then reflects the actual evidence gap, which is itself valuable signal (e.g., "no clinical trial data found for this drug in this indication").

Probe

5.

Is this built, or is this a prototype?

Core pipeline is fully functional: 4 live sources, LangGraph orchestration, Qdrant vector store, self-correction loop, and PDF report generation are all working. What remains is expanding source coverage (PubMed, DailyMed) and the evaluation benchmark.

Who is your target user?

Primary: clinical pharmacists, medical writers, drug safety teams in hospitals and pharma companies. Secondary: independent researchers and healthcare students who need fast, reliable drug dossiers. In India, this also addresses the massive under-resourced clinical pharmacy workforce.

Probe

2.

What's the size of this market?

Global drug intelligence / pharmacovigilance software market is valued above \$5B and growing 10%+ annually. Even targeting clinical decision support in Indian hospitals is a large addressable market given 1M+ registered pharmacists and 40,000+ hospitals.

Hard

4.

How is this different from just using ChatGPT or Perplexity?

ChatGPT has no structured access to FDA databases, ClinicalTrials.gov, or FAERS. Perplexity searches the open web, which includes unreliable sources. Aletheon pulls from authoritative, domain-specific APIs — and the evidence tier system means you know exactly how trustworthy each finding is. It's built for drug intelligence, not general Q&A.

Edge 1

Evidence trust tier system

Cortellis has curated data but no explicit trust hierarchy surfaced to the user. Aletheon assigns every piece of evidence a tier (Regulatory > Peer-reviewed > Real-world > Preprint > Patent) and surfaces this in the report. The user always

knows why a finding is high or low confidence. No competitor does this at the report level.

Edge 2

Self-correcting LangGraph pipeline

Cortellis added an AI Q&A layer in August 2025 — but it's a search layer on top of a static database. Aletheon's LangGraph orchestration actually runs fetch → combine → index → retrieve → evaluate → correct → critic as a closed loop. The model checks its own report against retrieved evidence and rewrites unsupported claims. No legacy platform does this.

Edge 3

Real-time multi-source aggregation (not a database)

Cortellis and Elsevier are databases — curated, updated periodically by human analysts. Aletheon fetches live from FDA, ClinicalTrials.gov, FAERS, and Europe PMC at query time. A drug approved yesterday or a trial registered this week is captured immediately. Cortellis would need an analyst to update their dataset first.

Edge 4

Accessibility and cost model

Cortellis starts at \$30,000+/year — inaccessible to independent researchers, students, hospital pharmacists, and small clinics. Aletheon is designed to work at the individual level, which opens up an entirely different market segment that these platforms structurally cannot serve.

Edge 5

Narrative report output vs. raw data

Cortellis, Elsevier, and most competitors give you a database to explore — you still do the synthesis work. Aletheon produces a finished, readable, cited report. The output is the product, not a data dashboard. This makes it usable by clinical pharmacists, medical writers, and students — not just data scientists.

"Cortellis just added AI in 2025. How are you different from that?"

Cortellis added a natural-language Q&A layer on top of their existing database — it's a search interface, not a synthesis engine. It answers questions about data that's already been manually curated. Aletheon fetches live from primary sources, synthesizes across them, and runs a self-correction loop on the output. The architecture is fundamentally different: theirs is query-over-database, ours is live-fetch-then-reason.

Hard

"Cortellis has 160+ analysts and 11 curated datasets. Your data quality can't match that."

Fair point — for deep pipeline analysis, competitive intelligence, and deal tracking, Cortellis is unmatched. We're not competing there. Aletheon is built for a different job: on-demand, per-drug evidence synthesis for clinical researchers and pharmacists who need an answer in minutes, not a \$30k subscription. Our sources — FDA labels, ClinicalTrials.gov, FAERS — are the same primary sources Cortellis uses. We're just making them accessible without the enterprise contract.

Hard

"What stops Clarivate or Elsevier from just building what you've built?"

Three things: (1) Legacy architecture — their are databases first. Rebuilding around live-fetch + LLM synthesis would cannibalize their curated-data

moat. (2) Incentive — their business model is annual enterprise contracts. A per-query, accessible tool threatens their pricing. (3) Speed — large companies with 160+ analysts and compliance requirements move slowly. We can ship and iterate in days. The startup advantage is exactly this window.

Probe

"Is this just a wrapper on top of public APIs?"

The data sources are public APIs — FDA labels, ClinicalTrials.gov, FAERS are all public. The value isn't the data, it's the architecture that sits on top: the Evidence normalization layer that makes 5 different API schemas comparable, the tier ranking system, the LangGraph self-correction loop, and the report generation pipeline. A "wrapper" implies trivial work. This is an orchestration and synthesis layer that required months of engineering to get right.

Probe

"Couldn't a researcher just use PubMed and read the papers themselves?"

Yes — that's exactly what they do today, and it takes 3–5 hours per drug. Aletheon does it in minutes with cross-source synthesis they couldn't easily do manually.

WHO THE COMMITTEE MIGHT COMPARE YOU TO

ALETHEON (YOURS) Open / grant-stage AI pipeline: fetch → tier-rank → synthesize → cite → PDF. Built for any researcher or clinician, not just big pharma. ✓ LLM synthesis + source grounding ✓ Evidence trust tiers ✓ Self-correction loop ✓ On-demand, per-drug reports ✓ Accessible to individuals	CLARIVATE CORTELLIS \$30,000+/year Curated enterprise database across 11+ modules. Manually maintained by 160+ analysts. Requires training to use. – AI added in Aug 2025 (Q&A layer only) ✗ No LLM synthesis ✗ No real-time FAERS aggregation ✓ Deep curated pipeline data ✗ Not affordable for individuals
ELSEVIER / RELX (GOLD STANDARD DD) Enterprise license Drug pricing + clinical content database. Structured data delivery — no synthesis, no narrative report output. ✗ No AI synthesis ✗ Focused on pricing/formulary ✓ High-quality clinical content ✗ No evidence tier system ✗ Export-oriented, not insight-oriented	PUBMED + CHATGPT (DIY) Free (but slow) What most researchers do today. Manual search across 5+ tabs, no cross-source synthesis, high hallucination risk. ✗ No source grounding ✗ No tier ranking ✗ Hours per drug ✗ No FAERS / FDA label pull ✓ Free, familiar

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More importantly, FAERS adverse event data is nearly impossible to manually synthesize — thousands of reports per drug. We surface the pattern; they'd need to read thousands of submissions themselves.